

LLM Output
(raw response)

Execution: L1 in every request | L2 sampled 5-10% | L3 in CI/CD and weekly

Level 1: Structural Validation (cheap, 100% coverage)

Schema Parsing
(JSON valid?
Pydantic model?)

Constraint Checks

- Length bounds
- Enum membership
- Cross-field logic
- No contradictions

✓ Pass / ✗ Reject
(immediate, <1ms)

passed structural
validation

Level 2: LLM-as-Judge (moderate cost, sampled or full)

Criteria Definition

- Relevance (1-5)
- Factual accuracy (1-5)
- Completeness (1-5)
- Tone/style (1-5)

Judge Model
(stronger model evaluates
weaker model's output)
GPT-5.5 judges Flash

Pairwise Comparison
(A vs B, run in BOTH
orders to detect
position bias)

scores feed into
statistical analysis

Level 3: Statistical Evaluation (expensive, periodic)

Golden Dataset
(curated test cases
w/ ground truth,
versioned)

Aggregate Metrics

- Mean score by category
- Score distribution
- Worst-case analysis
- p95 latency

Regression Detection
 $\Delta > \text{threshold?}$
→ Block deployment